

```
pip install numpy
```

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)

```
import numpy as np
print(np.__version__)
```

```
2.0.2
```

```
arr=np.array([1,2,3,4])
print(arr)
```

```
[1 2 3 4]
```

```
b=np.array([[1,2,3],
            [7,2,5]])
print(b)
```

```
[[1 2 3]
 [7 2 5]]
```

```
b=np.array([[1,2,3],
            [7,2,5],
            [1,2,3]])

print(b)
```

```
[[1 2 3]
 [7 2 5]
 [1 2 3]]
```

```
d=np.zeros([3,3])
print(d)
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

```
d=np.ones([3,3])
print(d)
```

```
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

```
d=np.full([3,3],8)
print(d)
```

```
[[8 8 8]
 [8 8 8]
 [8 8 8]]
```

```
d=np.eye(3)
print(d)
```

```
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
a=np.zeros([3,3])
b=np.ones([3,3])
c=np.full([3,3],8)
d=np.eye(3)
```

```
print(a,b,c,d)
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]] [[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
[[8 8 8]
 [8 8 8]
 [8 8 8]] [[1. 0. 0.]
```

```
[0. 1. 0.]
[0. 0. 1.]]
```

```
f=np.arange(2,10,3)
print(f)
```

```
[2 5 8]
```

```
np.linspace(0,1,5)
```

```
array([0. , 0.25, 0.5 , 0.75, 1.  ])
```

```
#uniform random
np.random.rand(3,3)
```

```
array([[0.60824725, 0.91269461, 0.01094423],
       [0.25920926, 0.97740339, 0.42951695],
       [0.99290393, 0.74986131, 0.90540509]])
```

```
#random integers
np.random.randint(1,5,size=(4,4))
```

```
array([[3, 2, 1, 1],
       [4, 2, 4, 4],
       [1, 3, 4, 3],
       [1, 3, 4, 4]])
```

```
#intialize grabage value
np.empty((2,4))
```

```
array([[3.53609937e-315, 0.00000000e+000, 2.30187695e+161,
        7.33723571e+223],
       [4.14114663e-114, 1.88719426e+122, 2.12152998e+088,
        1.23923386e+151]])
```

```
arr=np.array(["a","b","c","d","d","e"])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(1, 6)
2
6
<U1
4
24
```

```
a=np.array([10,20,30,40,50])
print(a[0])
print(a[2])
print(a[-1])
print(a[1:4:2])
print(a[::-2])
```

```
10
30
50
[20 40]
[10 30 50]
```

```
#slicing
print(arr[1:4])
print(arr[1:3])
print(arr[:2])
print(arr[::-1])
```

```
[]
[]
[['a' 'b' 'c' 'd' 'd' 'e']]
[['a' 'b' 'c' 'd' 'd' 'e']]
```

```
#2D indexing
m=np.array([[1,2,3],
            [2,3,4],
            [5,6,7]])
#single element(row,col)
print(m[0,0])
print(m[1,2])
#row slicing
print(m[0, :])
#column slicing
print(m[:, 1])
#sub-matrix
print(m[0:2,1:3])
```

```
1
4
[1 2 3]
[2 3 6]
[[2 3]
 [3 4]]
```

```
a=np.array([10,25,8,42,17])
#boolean mask
mask=a>15
print(a[mask])
#shorthand
print(a[a>5])
#fancy indexing
print(a[[2, 3]])
```

```
[25 42 17]
[10 25  8 42 17]
[ 8 42]
```

```
import numpy as np
```

```
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
print(a*b)
print(a/b)
print(a-b)
print(a**2)
print(a+10)
```

```
[5 7 9]
[ 4 10 18]
[0.25 0.4  0.5 ]
[-3 -3 -3]
[1 4 9]
[11 12 13]
```

```
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
print(a*b)
print(a/b)
print(a-b)
print(a**2)
print(a+10)
```

```
a=np.array([1,2,3,4])
np.ceil(a)
```

```
array([1.,  2.,  3.,  4.])
```

```
np.floor(a)
```

```
array([1.,  2.,  3.,  4.])
```

```
import numpy as np
```

```
arr=np.array([3,7,1,9,12])  
print(arr)
```

```
[ 3  7  1  9 12]
```

```
a=np.array([1,2,3])  
print(a)
```

```
[1 2 3]
```

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